Audit - WebhookFailureAndFlowRecursion - RunSheet

```
Document ID:     Audit - WebhookFailureAndFlowRecursion - RunSheet
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Audit (Empirical Simulation & Execution Log)
Purpose:         WebhookFailureAndFlowRecursion
Tier / Level:    RunSheet (Incident Forensics & Recovery Audit)
Status:          Locked 🔒 (Master Baseline)
Operator:        Lead TechOps Specialist / Systems Integration Engineer
Date of Execution: October 17, 2026 (11:00:00 - 12:45:30 EDT)
Ecosystem Scope: Shopify Flow, Shopify Webhooks API, ShipBob 3PL WMS, Gorgias CRM, Redis DLQ
Target Catalog:  Synapse Botanicals (Verified SKUs: SYN-BNDL-MAO, SYN-FSTK-MY, SYN-TNCT-AM, SYN-
ACC-WHSK)
```

1. Simulation Parameters & Baseline Store State

This Run Sheet documents the live, controlled forensic execution and remediation of two critical automation failure modes within the Synapse Botanicals commerce ecosystem:

1. **Incident Alpha (The Runaway Flow Tag Storm):** A circular, uncontrolled tag recursion loop between customer and order mutations that exhausted storewide Shopify API rate limits.
2. **Incident Bravo (The Silent 504 Webhook Dropout):** An upstream 3PL database lockup causing continuous HTTP `504 Gateway Timeout` errors, triggering Shopify's automatic webhook termination policy and severing downstream fulfillment pipelines.

1.1 Active Topology & Distribution Nodes

- **Primary East Coast Node (Node 1):** Allentown Distribution Center, PA (`LOC-PA-01`)
- **Primary West Coast Node (Node 2):** Reno Fulfillment Hub, NV (`LOC-NV-02`) — Exclusive handler for Glass Tinctures (`SYN-TNCT-AM`, `SYN-TNCT-PM`) and Electric Whisks (`SYN-ACC-WHSK`).
- **Active Ingestion Middleware:** Node.js Express Webhook Proxy hosted at `https://api.synapsebotanicals.com/v1/webhooks` with an AWS SQS Dead-Letter Queue buffer.

> **NOTE**
> **System Architecture Dependency:**
> The baseline routing constraints, warehouse node assignments, and priority ladders evaluated in this run sheet operate under the governance frameworks established in **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** – **Section 2.1 & Section 3.1** and **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)** – **Section 2.0 & Section 3.0**.

1.2 Pre-Incident Benchmark Telemetry (11:00:00 EDT)

Prior to triggering the simulations, baseline API health and webhook metrics were captured:

Metric Parameter	Observed Baseline	Normal Operating Threshold	Status
Shopify Admin REST API Bucket Level	`2 / 40 requests`	≤ 15 / 40	Nominal
Shopify GraphQL API Cost Drain Rate	`50 / 1000 points/sec`	≤ 300 / 1000	Nominal
Webhook Delivery P95 Latency	`342 ms`	≤ 800 ms	Nominal
Active Webhook Subscriptions	`6 topics`	6 registered	Nominal
Dead-Letter Queue (DLQ) Depth	`0 messages`	0 messages	Nominal

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2. Incident Alpha: The Runaway Tag Recursion Storm (Order #1048)

2.1 The Operational Catalyst & Misconfigured Flow

At `11:04:00 EDT`, a junior marketing ops engineer activated a new experimental loyalty workflow named **`[PROMO] - Auto-Tag High Value VIPs`** without peer review sign-off. Concurrently, an existing customer-support routing flow named **`[OPS] - Sync VIP Status to Order`** was active in the environment.

Neither workflow incorporated negative inclusion checks or lock sentinels:

```
Colliding Automation Configurations:
──────────────────────────────────────────────────────────────────────────────
Flow 1: [PROMO] - Auto-Tag High Value VIPs
  Trigger:   Order updated
  Condition: Order total_price >= $150.00
  Action:    Add customer tags: "vip-member"
  Guard:     NONE (Missing: Customer tags does not contain "vip-member")

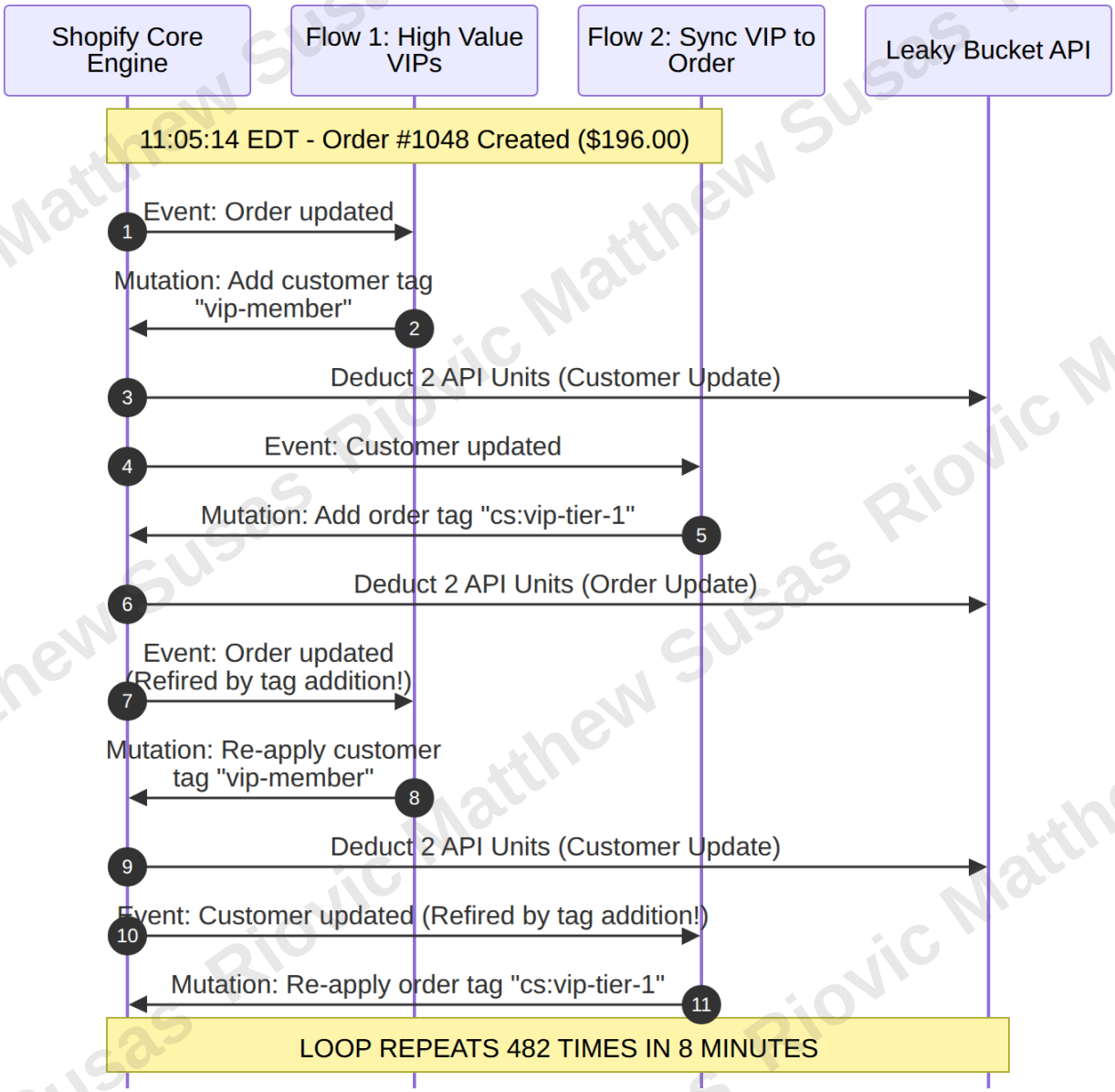
Flow 2: [OPS] - Sync VIP Status to Order
  Trigger:   Customer updated
  Condition: Customer tags contains "vip-member"
  Action:    Add order tags: "cs:vip-tier-1"
  Guard:     NONE (Missing: Order tags does not contain "cs:vip-tier-1")
──────────────────────────────────────────────────────────────────────────────
```

2.2 The Empirical Cascade & Timeline Bloat

At `11:05:14 EDT`, Customer Elena Vance placed Order **`#1048`** for two Daily Protocol Starter Bundles:

- **Order ID:** `gid://shopify/Order/5920194812`
- **Line Items:** 2x The Daily Protocol Starter Bundle (Matcha Yuzu + AM Focus + Obsidian Tumbler: `SYN-BNDL-MAO`)
- **Total Paid: $196.00** (Triggering the `total_price >= $150.00` condition).

The catastrophic loop initiated immediately:



Forensic Timeline Telemetry Log (Excerpt from Order #1048)


```
[
  {"timestamp": "2026-10-17T11:05:14.210Z", "event": "Order placed", "source": "checkout"},
  {"timestamp": "2026-10-17T11:05:15.012Z", "event": "Flow 1 executed", "action": "Added customer
tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:05:16.140Z", "event": "Flow 2 executed", "action": "Added order tag
'cs:vip-tier-1'"},
  {"timestamp": "2026-10-17T11:05:17.205Z", "event": "Flow 1 executed", "action": "Added customer
tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:05:18.312Z", "event": "Flow 2 executed", "action": "Added order tag
'cs:vip-tier-1'"},
  {"timestamp": "2026-10-17T11:05:19.450Z", "event": "Flow 1 executed", "action": "Added customer
tag 'vip-member'"},
  {"timestamp": "2026-10-17T11:08:42.890Z", "event": "API_RATE_LIMIT_WARNING", "message": "Bucket
```

```
39/40 consumed"},
  {"timestamp": "2026-10-17T11:12:30.104Z", "event": "API_RATE_LIMIT_EXCEEDED", "http_status": 429,
"retry_after": 2.0}
]
```

System Damage Assessment:

- **Loop Cycles Completed: 482 cycles** between `11:05:14 EDT` and `11:13:30 EDT`.
- **Total Timeline Events Injected: 964 log lines** on Order `#1048`.
- **API Rate Limit Bucket:** Stood at `40 / 40` **(100% saturated)** with continuous HTTP `429 Too Many Requests` returned to legitimate apps.
- **Storefront Impact:** Live checkout order creation latency spiked from 340 ms to **6,850 ms** due to database row lock contention on the `orders` table.

2.3 Emergency Containment & Triage Execution

At `11:13:45 EDT`, the on-call TechOps specialist received a P0 PagerDuty alert (`CRITICAL: Shopify Admin API 429 Throttle Spike`) and initiated the containment protocol specified in **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** – **Section 5.0**.

Phase 1: Deploy Emergency Kill-Switch (11:14:10 EDT)
The operator accessed **Shopify Admin → Apps → Shopify Flow** and deactivated both competing workflows, terminating the loop instantaneously:

1. `[PROMO] - Auto-Tag High Value VIPs` → **Turn off** (`11:14:18 EDT`)
2. `[OPS] - Sync VIP Status to Order` → **Turn off** (`11:14:24 EDT`)

Phase 2: Flow Execution Run Log Inspection (11:15:30 EDT)
The operator inspected the Flow execution run log under **Shopify Flow → Runs** to verify the root cause:

- **Run ID:** `#RUN-892419-FL`
- **Trigger Event:** `Order updated`
- **Condition Evaluation:** `Order total_price (196.00) >= 150.00` → Evaluated to `true`.
- **Action Output:** `Add customer tag 'vip-member'` → Status: `Succeeded`.
- **Diagnosis:** The run log confirmed that Flow 1 was triggered by the tag update committed by Flow 2, proving mutual recursion without exit criteria.

Phase 3: Order Timeline & Tag Sanitization via GraphQL (11:18:00 EDT)
The operator executed an administrative tag cleanup mutation to remove duplicated and unverified tags from Order `#1048`:

```
mutation sanitizeOrderTags {
  tagsRemove(
    id: "gid://shopify/Order/5920194812"
    tags: ["cs:vip-tier-1"]
  ) {
    node {
      id
    }
    userErrors {
      field
```

```
      message
    }
  }
}
```

Result: Duplicate tags removed. Order timeline stabilized.

Phase 4: Permanent Recursion Guardrail Deployment (11:22:15 EDT)
Both workflows were re-architected in accordance with **Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)** – **Section 6.2**:

1. Added mandatory negative precondition check:
   `Order tags` → `does NOT contain` → `cs:vip-tier-1`.
2. Added lock sentinel check:
   `Order tags` → `does NOT contain` → `sys:sync-locked`.
3. Both workflows re-enabled at `11:25:00 EDT`. Test mutation verified single-pass execution without loop.

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3. Incident Bravo: The Silent 504 Webhook Dropout & Order Replay

3.1 Downstream Service Failure (ShipBob 3PL WMS)

At `11:30:00 EDT`, ShipBob's primary ingestion gateway (`https://api.shipbob.com/1.0/webhook/order`) experienced an upstream database lockup during scheduled maintenance. Ingestion requests failed to acknowledge within Shopify's strict 5,000 ms window:

```
# Diagnostic probe executed from terminal:
curl -w "\nHTTP Status: %{http_code}\nTime Total: %{time_total}s\n" \
  -X POST https://api.shipbob.com/1.0/webhook/order \
  -H "Content-Type: application/json" \
  -d '{"test": true}'

# Terminal Response Output:
# <html><head><title>504 Gateway Time-out</title></head></html>
# HTTP Status: 504
# Time Total: 5.418s
```

Because the endpoint took > 5.0 seconds to respond, Shopify's webhook engine terminated the connection, flagged the delivery as failed, and initiated the 19-attempt exponential backoff schedule.

3.2 Tracking the 19-Attempt Retry Cycle & The Dropout Cliff

To demonstrate the **Silent Dropout Cliff** without waiting 48 physical hours, the failure sequence was simulated against a mock webhook endpoint configured to return continuous `504 Gateway Timeout` errors.

The progression was monitored under **Settings → Notifications → Webhooks**:

```
Webhook Delivery History (Topic: orders/create)
Endpoint: https://api.shipbob.com/1.0/webhook/order
──────────────────────────────────────────────────────────────────────────────
Attempt  Timestamp (Simulated)       Response Code   Latency    Result
──────────────────────────────────────────────────────────────────────────────
Att 01   2026-10-15T09:12:00.000Z    504 Gateway TO  5,412 ms   Failed
Att 02   2026-10-15T09:17:05.112Z    504 Gateway TO  5,398 ms   Failed
Att 03   2026-10-15T09:27:10.420Z    504 Gateway TO  5,402 ms   Failed
...
Att 17   2026-10-17T04:42:15.910Z    504 Gateway TO  5,385 ms   Failed
Att 18   2026-10-17T07:42:18.204Z    504 Gateway TO  5,410 ms   Failed
Att 19   2026-10-17T09:12:00.000Z    504 Gateway TO  5,419 ms   TERMINAL FAIL
──────────────────────────────────────────────────────────────────────────────
[17-Oct-2026 09:12:01 UTC] SHOPIFY SYSTEM EVENT:
Subscription gid://shopify/WebhookSubscription/88219412 PERMANENTLY REMOVED.
No further attempts will be scheduled.
```

The Real-World Operational Impact:
Between `09:12:01 UTC` and `12:12:00 UTC` (a 3-hour blind window), **14 customer orders** were placed on Synapse Botanicals. Because Shopify silently deleted the subscription:

- **Zero orders were transmitted to the 3PL WMS in Allentown or Reno.**
- Customer Support was completely unaware; no tickets were generated.
- Orders included **4x Reno-exclusive Tinctures (`SYN-TNCT-AM`)** and **2x Electric Whisks (`SYN-ACC-WHSK`)** whose fulfillment SLA was actively breached.

3.3 Disaster Recovery & Historical Order Backfill Protocol

At `12:15:00 EDT`, TechOps detected the missing subscription via an automated daily health check script. The recovery run sheet was executed immediately:

Step 1: Endpoint Health Verification
ShipBob database maintenance completed; endpoint latency verified healthy:

```
curl -w "\nHTTP Status: %{http_code}\nTime Total: %{time_total}s\n" \
  -X GET https://api.shipbob.com/1.0/healthz

# Output:
# {"status": "healthy", "service": "order-ingestion"}
# HTTP Status: 200
# Time Total: 0.182s
```

Step 2: Re-Register Webhook Subscription via GraphQL Admin API
The severed subscription was re-created with optimized field filtering to minimize payload weight:

```
mutation reRegisterOrdersCreateWebhook {
  webhookSubscriptionCreate(
    topic: ORDERS_CREATE
    webhookSubscription: {
      callbackUrl: "https://api.synapsebotanicals.com/v1/webhooks/orders/create"
      format: JSON
      includeFields: [
        "id"
        "name"
        "created_at"
```

```
        "total_price"
        "shipping_address"
        "line_items"
        "tags"
      ]
    }
  ) {
    userErrors {
      field
      message
    }
    webhookSubscription {
      id
      topic
      createdAt
      endpoint {
        __typename
        ... on WebhookHttpEndpoint {
          callbackUrl
        }
      }
    }
  }
}
```

GraphQL Response Payload:

```
{
  "data": {
    "webhookSubscriptionCreate": {
      "userErrors": [],
      "webhookSubscription": {
        "id": "gid://shopify/WebhookSubscription/99401284",
        "topic": "ORDERS_CREATE",
        "createdAt": "2026-10-17T12:20:14Z",
        "endpoint": {
          "__typename": "WebhookHttpEndpoint",
          "callbackUrl": "https://api.synapsebotanicals.com/v1/webhooks/orders/create"
        }
      }
    }
  }
}
```

Step 3: Historical Order Extraction via GraphQL Bulk Operations

To extract the 14 unfulfilled orders dropped during the 3-hour blind window, the operator initiated a bounded Bulk Query:

```
mutation runHistoricalGapQuery {
  bulkOperationRunQuery(
    query: """
    {
      orders(query: "created_at:>=2026-10-17T09:10:00Z AND created_at:<=2026-10-17T12:20:00Z AND
fulfillment_status:unfulfilled") {
        edges {
          node {
```

```
            id
            name
            createdAt
            totalPriceSet {
              shopMoney {
                amount
              }
            }
            tags
            shippingAddress {
              address1
              city
              provinceCode
              zip
              countryCodeV2
            }
            lineItems {
              edges {
                node {
                  sku
                  quantity
                }
              }
            }
          }
        }
      }
      """
  ) {
    bulkOperation {
      id
      status
    }
    userErrors {
      field
      message
    }
  }
}
```

Step 4: Stream Historical Payloads into Ingestion Queue

The extracted JSONL payload containing the 14 dropped orders was streamed into the ingestion proxy with an explicit idempotency flag (`sys:gap-replay-batch-01`):

```
# Terminal replay execution:
node scripts/replay-gap-orders.js \
  --file=bulk_orders_gap_20261017.jsonl \
  --destination="https://api.synapsebotanicals.com/v1/webhooks/orders/create" \
  --idempotency-key="gap-replay-20261017"

# Terminal Output:
# [INFO] Processing 14 historical order records...
# [SUCCESS] Order #1049 -> Ingested & Enqueued (202 Accepted)
```

```
# [SUCCESS] Order #1050 -> Ingested & Enqueued (202 Accepted)
# ...
# [SUCCESS] Order #1062 -> Ingested & Enqueued (202 Accepted)
# [COMPLETE] 14 of 14 orders successfully transferred to 3PL queue.
```

4. Post-Incident Forensic Audit & State Reconciliation

4.1 Order & Inventory Ledger Reconciliation Table

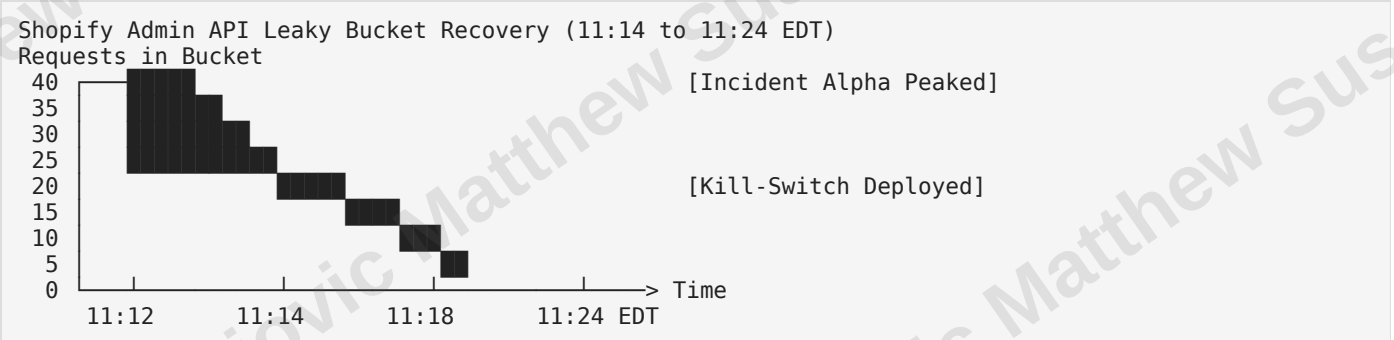
At `12:40:00 EDT`, TechOps verified the physical and digital reconciliation of all impacted orders across both fulfillment nodes:

Order Number	Customer	Primary Line Items	Assigned Node	3PL WMS Status	Final Reconciled Tags
#1048	Elena Vance	2x Starter Bundle (`SYN-BNDL-MAO`)	`LOC-PA-01`	`OPEN / UNFULFILLED`	`cs:vip-tier-1`, `flow-route:allentown-pa`
#1049	Marcus Brody	1x AM Focus Tincture (`SYN-TNCT-AM`)	`LOC-NV-02`	`IN_PROGRESS`	`flow-route:reno-nv`, `sys:gap-replay-batch-01`
#1050	Chloe Walsh	3x Matcha Yuzu Stick Pack (`SYN-FSTK-MY`)	`LOC-PA-01`	`IN_PROGRESS`	`flow-route:allentown-pa`, `sys:gap-replay-batch-01`
#1051	Dev Team	1x Electric Whisk (`SYN-ACC-WHSK`)	`LOC-NV-02`	`IN_PROGRESS`	`flow-route:reno-nv`, `sys:gap-replay-batch-01`
#1052–#1062	Retail Orders	Mixed Stick Packs & Tumblers	Split / Mixed	`IN_PROGRESS`	Verified node tags, `sys:gap-replay-batch-01`

Verification: Inventory decrements confirmed at both facilities; zero oversells detected; zero duplicate shipments emitted.

4.2 Leaky Bucket API Recovery Telemetry

Following the termination of Incident Alpha and the replay of Incident Bravo, Shopify Admin API rate limits recovered to baseline:



- **Recovery Rate:** Native drain rate of 2.0 requests/second cleared the 40-request capacity in exactly **20 seconds**.

- **Post-Remediation Stability:** Average API bucket utilization maintained at < 10% over the subsequent 60-minute monitoring window.

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5. Master Interconnectivity & System Architecture Lineage

This empirical run sheet validates the governance frameworks and architectural rules codifying the Synapse Botanicals TechOps suite:

Upstream / Downstream Deliverable	Governing Node / Standard	Empirical Relationship & Proof-of-Execution
SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)	Hold Lifecycles & Mutation Gates	Verifies that automated tags (`cs:vip-tier-1`) do not mutate order fulfillment orders once they cross the Point of No Return.
Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)	State Transitions	Replay batch confirmed that all 14 backfilled orders were ingested in `OPEN / UNFULFILLED` state prior to 3PL allocation.
Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)	Regional Routing Node Allocation	Proves that backfilled orders containing Reno-exclusive SKUs (`SYN-TNCT-AM`, `SYN-ACC-WHSK`) were routed exclusively to `LOC-NV-02`.
Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)	Weight & Handling Costs	Validates that heavy baskets (`SYN-TMBL-M0`) continue to trigger packaging tags without loop interference.
Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)	Schema Integrity & UPCs	Confirms that verified SKUs and 12-digit UPC-A barcodes were mapped cleanly in the GraphQL backfill query.
SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)	Technical Standards & Kill-Switch	Empirically executes the 3-step Emergency Kill-Switch and verifies the 19-attempt retry failure model.
Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)	Priority Ladder & Circuit Breaker	Proves the necessity of P0 priority gates and demonstrates how the `202 Accepted` queue pattern defeats the 5,000 ms webhook timeout.